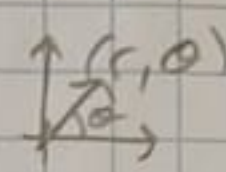
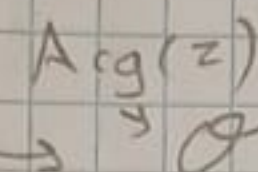


Division: $\frac{z_1}{z_2} = \frac{(x_1 x_2 + y_1 y_2) + i(x_2 y_1 - x_1 y_2)}{x_2^2 + y_2^2}$

Conjugate: $z = a + ib \Rightarrow \bar{z} = a - ib$

Modulus: $|z| = \sqrt{x^2 + y^2} \Rightarrow \text{length from } (0,0) \text{ to } (x,y)$

Polar form: $z = r(\cos \theta + i \sin \theta)$, $r = |z|$ 

Argument: $-\pi \leq \text{Arg}(z) \leq \pi \rightarrow \text{Arg}(z)$  θ açısını verir

$\arg(z) = \text{Arg}(z) + 2n\pi$ $\arg(z)$, sınırlanmış değildir

Multiplication: $z_1 \cdot z_2 = r_1 \cdot r_2 (\cos(\theta_1 + \theta_2) + i \sin(\theta_1 + \theta_2))$

Division: $z_1/z_2 = \frac{r_1}{r_2} [\cos(\theta_1 - \theta_2) + i \sin(\theta_1 - \theta_2)]$

$\arg(z_1 z_2) = \arg(z_1) + \arg(z_2)$

$\arg(z_1/z_2) = \arg(z_1) - \arg(z_2)$

$z^n = r^n [\cos(n\theta) + i \sin(n\theta)]$

$z^{1/n} = \sqrt[n]{z} = w = \sqrt[n]{r} \left[\cos\left(\frac{\theta}{n} + \frac{2k\pi}{n}\right) + i \sin\left(\frac{\theta}{n} + \frac{2k\pi}{n}\right) \right]$, $k=0, 1, \dots, n-1$

$k=0 \Rightarrow z^{1/n} = \dots$

$k=1 \Rightarrow \dots$

$k=2 \Rightarrow \dots$

} bu şekilde sonuçlar yazılır

Circles: $|z - z_0| = p$, $p > 0$ then z lie on the circle of radius p centered at z_0 .

$e^z = e^{x+iy} = e^x (\cos y + i \sin y) = e^x \cos y + e^x i \sin y$

$z = r e^{i\theta} \Rightarrow z^{1/n} = \sqrt[n]{r} e^{i(\frac{\theta}{n} + \frac{2k\pi}{n})}$

complex expo. function

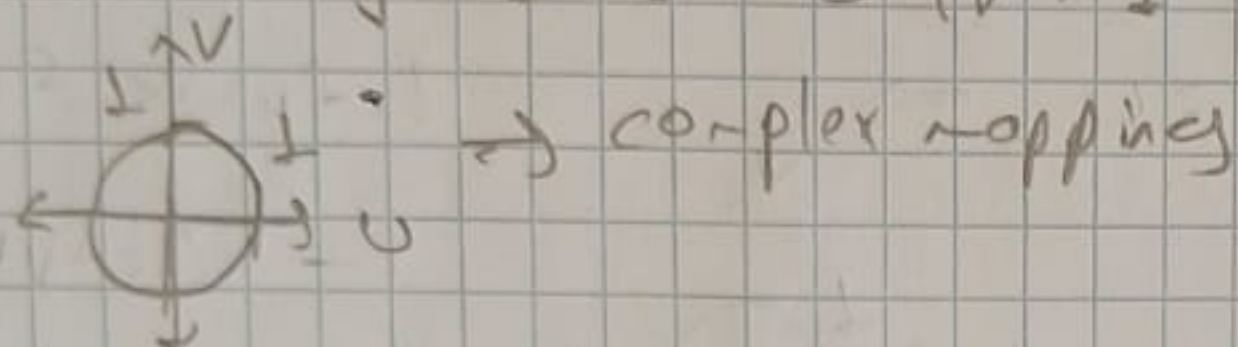
2

Complex mapping: a) find ^{image of} $|z|=1$ under $w = \frac{1}{z}$

Sol a) $z = x+iy \Rightarrow |z| = \sqrt{x^2+y^2} = 1 \Rightarrow x^2+y^2=1$

$w = \frac{1}{z} = \frac{1}{x+iy} = \frac{x-iy}{x^2+y^2} = x-iy \Rightarrow u(x,y)=x, v(x,y)=-y$

$u^2+v^2 = x^2+y^2 = 1 \Rightarrow u^2+v^2=1 \rightarrow \text{circle}$



Curves

Line: $z(t) = z_0 + t(z_1 - z_0), -\infty < t < \infty$

Line Segment: $0 \leq t \leq 1$

Ray: $0 \leq t < \infty$

Circle: $z(t) = z_0 + r(\cos t + i \sin t), 0 \leq t \leq 2\pi$

center: z_0 radius: $r, z(t) = z_0 + r e^{it}$

Image of curve $z(t)$ under $w = F(z)$: $w(t), a \leq t \leq b$

Image: $w(t) = F(z(t)), a \leq t \leq b$

Limit: $\lim_{z \rightarrow z_0} F(z) = L$ if for every $\epsilon > 0$ there exist a $\delta > 0$ such that $|F(z) - L| < \epsilon$ whenever $0 < |z - z_0| < \delta$

By asking to prove dense Kullonirz.

Ex: Show $\lim_{z \rightarrow 0} \frac{z}{z}$ does not exist

$\lim_{z \rightarrow 0} \frac{x+0i}{x-0i} = 1 = L_1, \lim_{z \rightarrow 0} \frac{0+i}{0-i} = -1 = L_2, L_1 \neq L_2$ Limit doesn't exist

* $F(z) = u + iv, z_0 = x + iy, \lim_{z \rightarrow z_0} F(z) = L$

$L = u_0 + i v_0 \leftarrow \lim_{(x,y) \rightarrow (x_0,y_0)} u(x,y) = u_0$

$\lim_{(x,y) \rightarrow (x_0,y_0)} v(x,y) = v_0$

Continuity: i) $\lim_{z \rightarrow z_0} f(z)$ exists

ii) $f(z)$ is defined at z_0 , iii) $\lim_{z \rightarrow z_0} f(z) = f(z_0)$

$f = u + iv$ is continuous at z_0 if both u and v are continuous at z_0 (x_0, y_0)

Cauchy-Riemann: $f(z) = u + iv$, $f'(z)$ exist

$$* \frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}, \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$$

$$* \frac{\partial u}{\partial r} = \frac{1}{r} \frac{\partial v}{\partial \theta}, \quad \frac{\partial v}{\partial r} = -\frac{1}{r} \frac{\partial u}{\partial \theta}$$

$$\text{Derivative: } f'(z) = \frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x} = \frac{\partial v}{\partial y} - i \frac{\partial u}{\partial y}$$

$$f'(z) = \frac{1}{e^{i\theta}} \left(\frac{\partial u}{\partial r} + i \frac{\partial v}{\partial r} \right) = \frac{1}{r} \cdot \frac{1}{e^{i\theta}} \left(\frac{\partial v}{\partial \theta} - i \frac{\partial u}{\partial \theta} \right)$$

Analytic Function: Sayısı sıfır olan yerler hariç tüm polinomlar analitiktir.

$f(z)$, z_0 ve komşuluklarında türemlenebilir ve Cauchy-Riemann sağlıyorsa fonksiyon analitiktir.

$$\text{Harmonic Function: } \nabla^2 \phi = \frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} = 0$$

Harmonic Conjugate Function: $U(x, y)$ harmonic olsun, eğer u ile Cauchy-Riemann koşullarını sağlayan bir $v(x, y)$ varsa $v(x, y)$ is harmonic conjugate of $u(x, y)$

Sorularda $U(x, y)$ verilir $\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}$ ve $\frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$ eşitlikleri ile $v(x, y)$ 'yi bulunuz.

④

$$e^z = e^x (\cos y + i \sin y)$$

her yerde
analitik
 e^z is entire function

$$|e^z| = e^x, \quad \text{Arg}(e^z) = y + 2n\pi, \quad \overline{e^z} = e^{\bar{z}}$$

Periodicity: $F(z+T) = F(z), \quad T = 2\pi i \text{ for } e^z$